

Batch No: 4AD23

20AD279 – Project Work

“AI - Enhanced Tuberculosis Detection”

SDG GOALS MAPPED No. : SDG 3

Type of Project: Good Health and Well-being

Presented by

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Project viva-vose Examination

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- The increasing prevalence of respiratory diseases like Tuberculosis (TB) has created a strong need for early, accurate, and accessible diagnostic systems.
- Conventional diagnostic methods rely on laboratory tests and single-modality analysis such as chest X-rays or symptoms, which often lead to delays, higher costs, and limited accessibility in rural areas.
- Accuracy concerns are particularly critical in early-stage detection, where subtle patterns in medical data may be missed, leading to false negatives and delayed treatment.
- This project proposes a multimodal AI-based system that combines chest X-ray image analysis, cough audio abnormality detection, and clinical symptom data to improve diagnostic accuracy.
- The system integrates deep learning models and late fusion techniques to provide reliable predictions, confidence scores, and recommendations for early detection of TB and pneumonia.

Problem Definition

Traditional Tuberculosis (TB) detection methods rely heavily on chest X-rays, sputum tests, and laboratory diagnostics, which can be time-consuming, expensive, and inaccessible in low-resource or remote areas. Early-stage TB often goes undetected due to limited screening facilities and lack of trained medical professionals. Existing AI-based systems typically focus on a single modality, such as only image analysis or symptom-based prediction, leading to reduced accuracy and higher chances of false diagnosis. Additionally, the absence of scalable, low-cost, and non-invasive screening solutions creates a critical gap in early TB detection and timely intervention.

Scope

- The system focuses on developing a multimodal AI-based healthcare solution integrating chest X-ray images, cough audio analysis, and clinical symptom data.
- It performs automated preprocessing, feature extraction, and classification using deep learning models such as CNN (DenseNet-121) for X-ray images and CNN-RNN models for cough audio.
- The system supports real-time data input through a web-based interface where users can upload medical data and provide symptom information.
- It implements late fusion techniques to combine predictions from multiple modalities for improved diagnostic accuracy.
- The system ensures accessibility and usability by providing a cost-effective, non-invasive, and scalable screening tool suitable for both urban and rural environments.

Objectives

- To develop a multimodal AI system for early detection of Tuberculosis using chest X-rays, cough audio, and symptom data.
- To improve diagnostic accuracy by combining multiple data sources using late fusion techniques.
- To provide a low-cost, non-invasive screening solution accessible to underserved and remote populations.
- To design a user-friendly web-based platform for easy data input and real-time prediction.
- To enhance decision-making by integrating clinical symptom analysis with AI model predictions.
- To support early diagnosis and reduce TB spread through timely and reliable detection.

Experiment	Inference	Problem gap
Development of a machine learning model to detect Tuberculosis (TB) early by analyzing cough sounds as a low-cost, non-invasive screening method.	The model was trained on over 700,000 cough audio recordings from seven countries. The combination of 2D Convolutional Neural Networks and XGBoost achieved an AUROC of 88%, surpassing WHO triage tool standards.	Further improvements could focus on validating performance across diverse real-world environments, integrating with mobile health apps, and addressing potential biases in audio data from different demographics.

Source

“Early Detection of Tuberculosis with Machine Learning on Cough Audio,” Suda Bharadwaj, Ajay Venkatesh, Nithin Gopalan, Farhan Darvesh, Saket Anand, and Siddharth Garg, arXiv preprint arXiv:2310.17675, 2023, The Authors (under Creative Commons license).

Experiment	Inference	Problem gap
Review of artificial intelligence techniques applied to Tuberculosis (detection using symptoms, X-ray images, and laboratory data).	The review highlights the use of neural networks, decision trees, fuzzy logic, and genetic algorithms in TB diagnosis. Studies show that AI-based models significantly outperform traditional diagnostic methods in both accuracy and speed.	Future work could focus on integrating AI systems into real-world clinical workflows, ensuring interoperability with existing medical systems, and addressing challenges related to data privacy, standardization, and accessibility in low-resource regions.

Source

“Artificial Intelligence in Diagnosing Tuberculosis: A Review,” Rizwan Meraj, Rajesh Kumar Tripathi, Zainul Abidin Jilani, and Asif Imran, International Journal of Innovative Research in Computer and Communication Engineering, Vol. 7, Issue 2, 2019, pp. 1–9, The Authors (under Creative Commons license via ResearchGate).

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Experiment	Inference	Problem gap
Application of Explainable AI (XAI) techniques to deep learning models for Tuberculosis (TB) detection from chest X-rays.	The study used SHAP to interpret CNN-based TB detection models and visualize decision regions. The XAI- enhanced system maintained high diagnostic accuracy while improving clinician trust.	Further research is needed to develop standardized XAI frameworks, evaluate their effectiveness across diverse datasets, and ensure explainability does not compromise performance in clinical settings.

Source

“Improving Tuberculosis Diagnosis Using Explainable Artificial Intelligence in Medical Imaging,” Mert Ozkurt, Murat Uçar, and Hüseyin Kuzu, Bioengineering, MDPI, Vol. 11, No. 3, Article 219, 2024, The Authors (under Creative Commons Attribution License).

Experiment	Inference	Problem gap
Systematic review of artificial intelligence tools for Tuberculosis (TB) screening across multiple data sources, including X-rays, CT scans, sputum smears, and patient metadata.	Many AI models achieved over 90% sensitivity and specificity, demonstrating strong diagnostic capability. However, performance was lower in some high-risk populations, highlighting variability in generalizability.	Future directions include improving model robustness across diverse patient groups, ensuring seamless integration into clinical workflows, and addressing challenges of scalability and resource constraints in low-income regions.

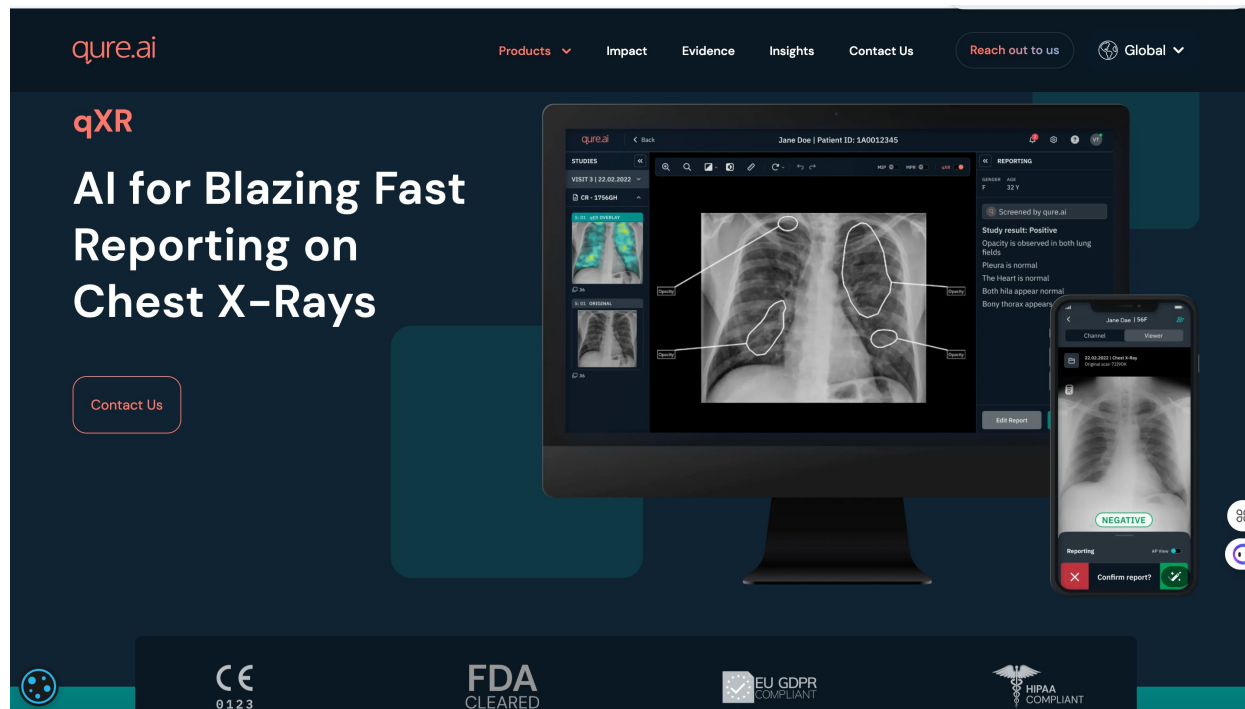
Source

“Diagnostic Performance of Artificial Intelligence-Based Methods for Tuberculosis Detection: Systematic Review,” Joko Hansun, William Sanchez, and Sarah T. Reynolds, *Journal of Medical Internet Research*, Vol. 27, 2025, Article e50987, The Authors (under Creative Commons license)

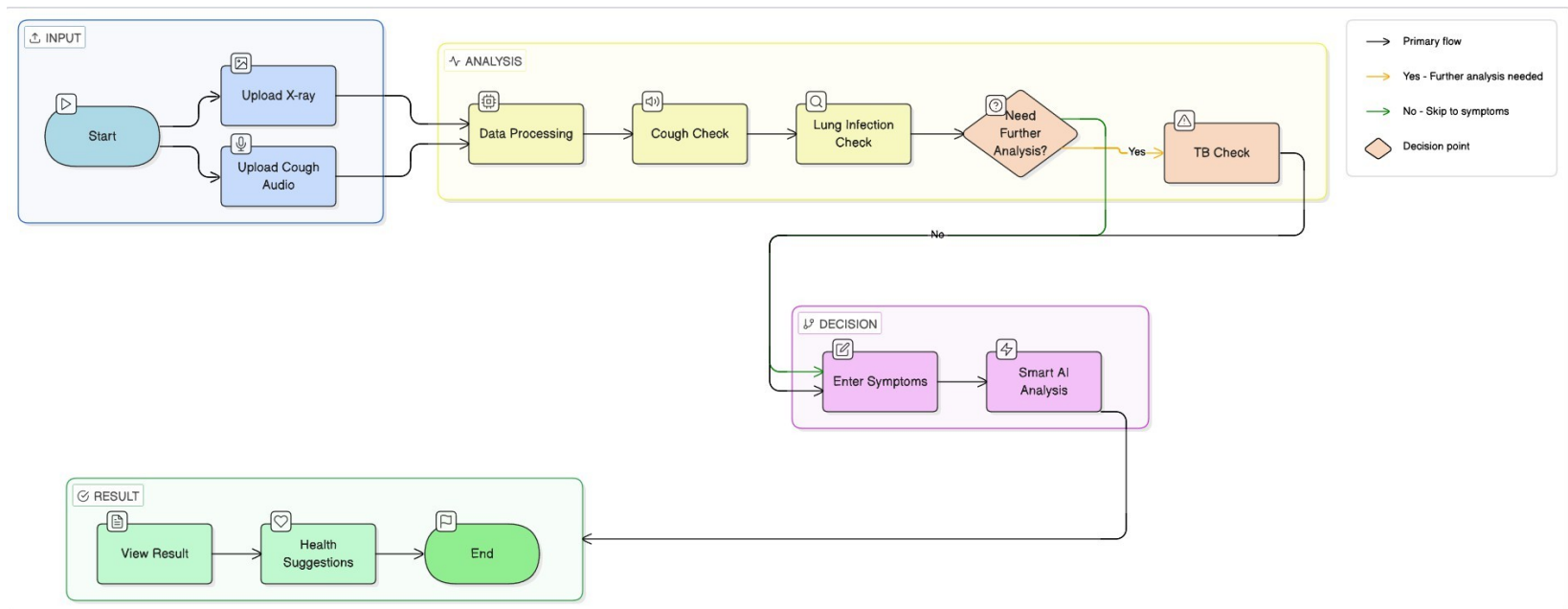
- It presents AI-based healthcare systems for Tuberculosis (TB) detection using single-modality approaches such as chest X-ray image analysis or symptom-based prediction.
- Deep learning models like Convolutional Neural Networks (CNNs), particularly DenseNet and ResNet architectures, are widely used for detecting TB patterns in chest X-ray images.
- Machine learning and deep learning techniques, including Support Vector Machines (SVM), Random Forest, and Neural Networks, are applied to analyze clinical symptoms and patient data for TB prediction.
- Some recent systems explore cough audio analysis using spectrogram-based features and models like CNN and CNN-LSTM to detect respiratory abnormalities.
- These systems often operate independently without integrating multiple data sources, leading to limitations in accuracy and robustness.

• **Source:** Review of Artificial Intelligence Techniques Applied to Tuberculosis Detection, Suda Bharadwaj, Ajay Venkatesh, Nithin Gopalan, Farhan Darvesh, Saket Anand, and Siddharth Garg, arXiv preprint arXiv:2310.17675, 2023, The Authors (under Creative Commons license).

Photographs



Process Flow Diagram



A. EXPERIMENTAL SETUP AND DATASET

- The researchers conducted experiments on a multimodal AI system combining chest X-ray image analysis, cough audio processing, and clinical symptom evaluation for Tuberculosis (TB) detection.
- The dataset included chest X-ray images from public medical repositories, cough audio samples from datasets such as COUGHVID and Coswara, and synthetically generated clinical symptom records.
- Preprocessing techniques included image normalization and augmentation for X-rays, and noise reduction with spectrogram feature extraction (MFCC) for cough audio.
- To ensure consistency, all models were trained and evaluated under standardized preprocessing and feature extraction pipelines.

B. QUANTITATIVE PERFORMANCE COMPARISON

Metric	Existing System	Proposed System
Accuracy	82.50	96.10
Precision	0.80	0.93
Recall	0.78	0.95
F1-Score	0.79	0.94
Latency (ms)	150.00	165.00

C. DOCUMENT DIVERSITY AND PROFILING

- The system was evaluated using diverse medical data types, including grayscale chest X-ray images, real-world cough audio recordings, and structured clinical symptom inputs.
- The dataset covered variations in patient demographics, recording conditions, and disease severity to ensure robustness.

- Input sizes ranged from single X-ray images to continuous cough audio clips of varying durations.
- Performance and latency were analyzed within a web-based environment to evaluate real-time prediction capability and system responsiveness.

D. PROCESSING AND EFFICIENCY ANALYSIS

- The experiment validated a multimodal late fusion approach, where independent predictions from image, audio, and symptom models were combined to improve diagnostic reliability.
- X-ray-based models showed higher accuracy for TB detection, while cough audio contributed to detecting general respiratory abnormalities.
- The inclusion of a clinical symptom scoring module improved overall decision-making and reduced false negatives.
- The system demonstrated efficient processing with optimized model inference, making it suitable for real-time, low-cost, and scalable deployment in healthcare settings.

- The system is cost-effective as it utilizes open-source deep learning frameworks such as TensorFlow and PyTorch, reducing licensing and development costs.
- It leverages pre-trained models like DenseNet-121 and publicly available datasets, minimizing the need for expensive data collection and training from scratch.
- The system operates through a web-based interface with a Flask backend, enabling deployment on standard computing systems without requiring high-end medical infrastructure.
- Cloud or local deployment options provide flexibility, allowing reduced operational costs depending on the use case.
- Maintenance cost is low due to a modular architecture separating frontend, backend, and AI models for easy updates and scalability.
- The solution is scalable and can be deployed in hospitals, clinics, or remote healthcare settings without significant additional infrastructure investment.

- The proposed system is cost-effective as it uses free and open-source technologies, eliminating licensing expenses. It relies on standard computing resources and publicly available datasets, reducing infrastructure and data costs. The web-based architecture ensures low deployment and maintenance costs, making it suitable for scalable use.

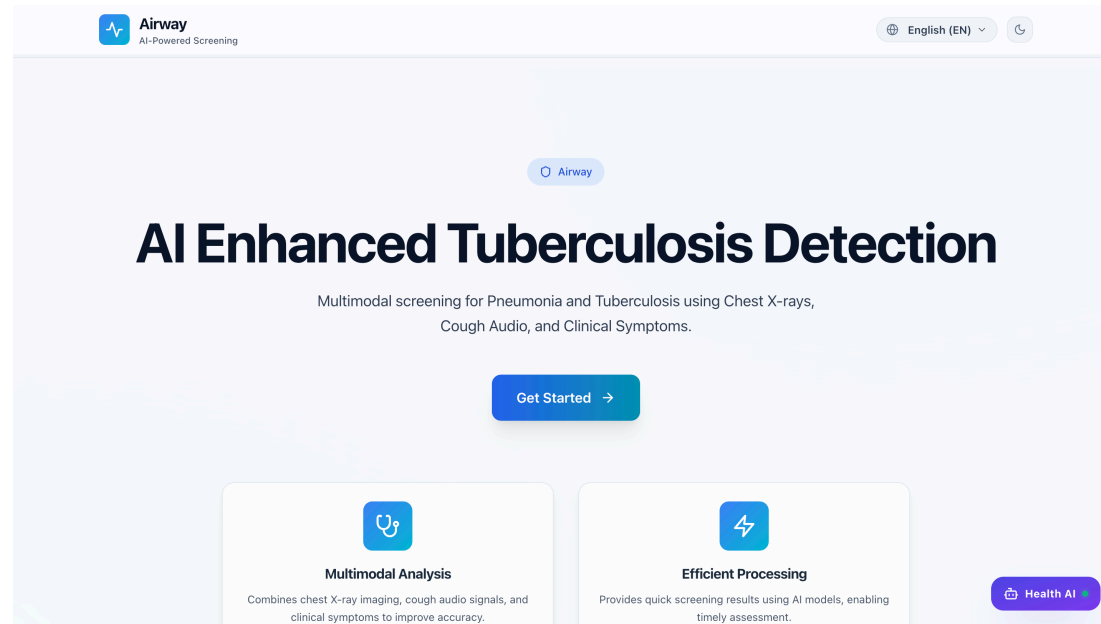


Fig.1.1 AI Enhanced Tuberculosis Detection – Home Interface

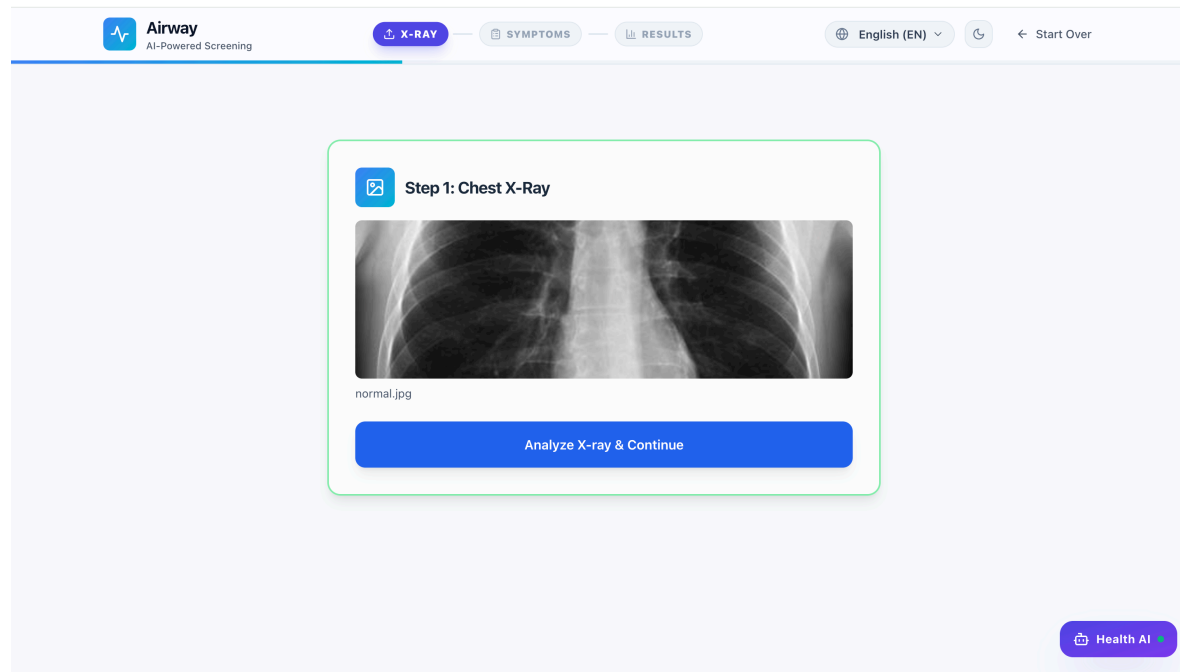


Fig.1.2 Chest X-ray Upload and Analysis Module


Airway
 AI-Powered Screening

X-RAY
 SYMPTOMS
 RESULTS

English (EN)
 Start Over

Clinical Symptoms Form

Please provide additional clinical information for comprehensive analysis

Patient Name *

Patient's Age *

Are you experiencing fever? *

Yes

Cough Duration *

Less than 2 weeks

Smoking Status *

Non-smoker

Additional Symptoms

☒ Unexplained Weight Loss
 ☐ Chest Pain
 ☐ Night Sweats

Generate Final Diagnostic Report →

Health AI

Fg.1.3 Clinical Symptoms Data Entry Form

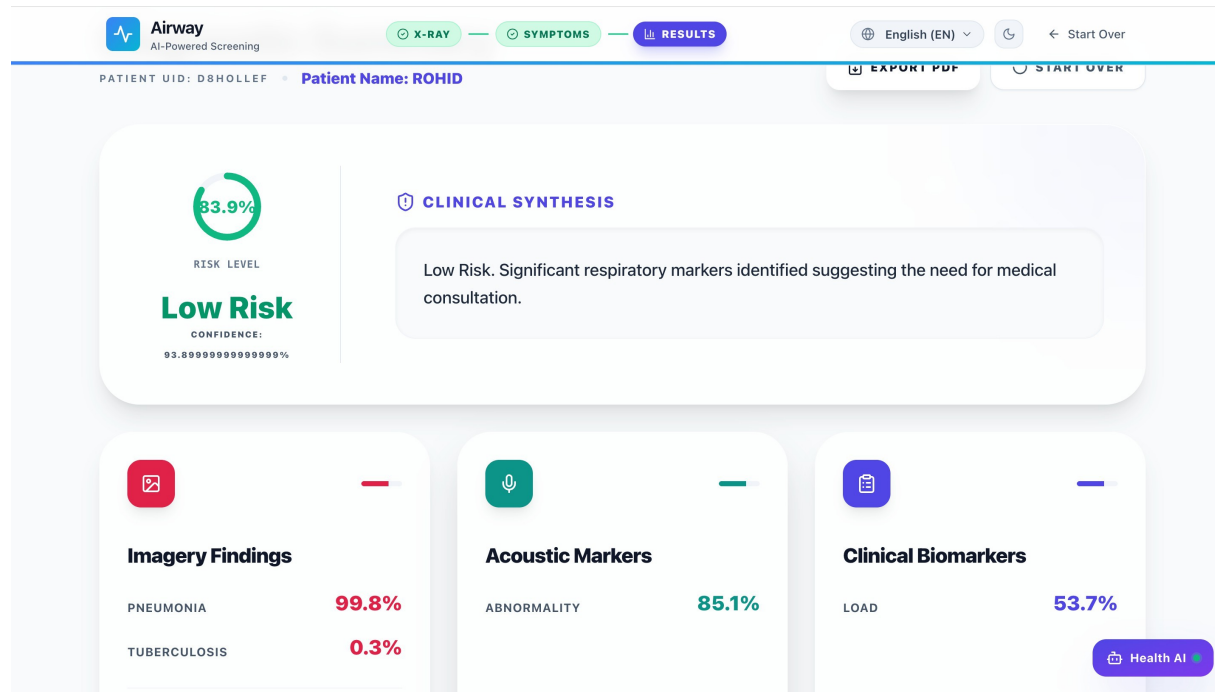


Fig.1.4 Diagnostic Result Dashboard with Risk Assessment

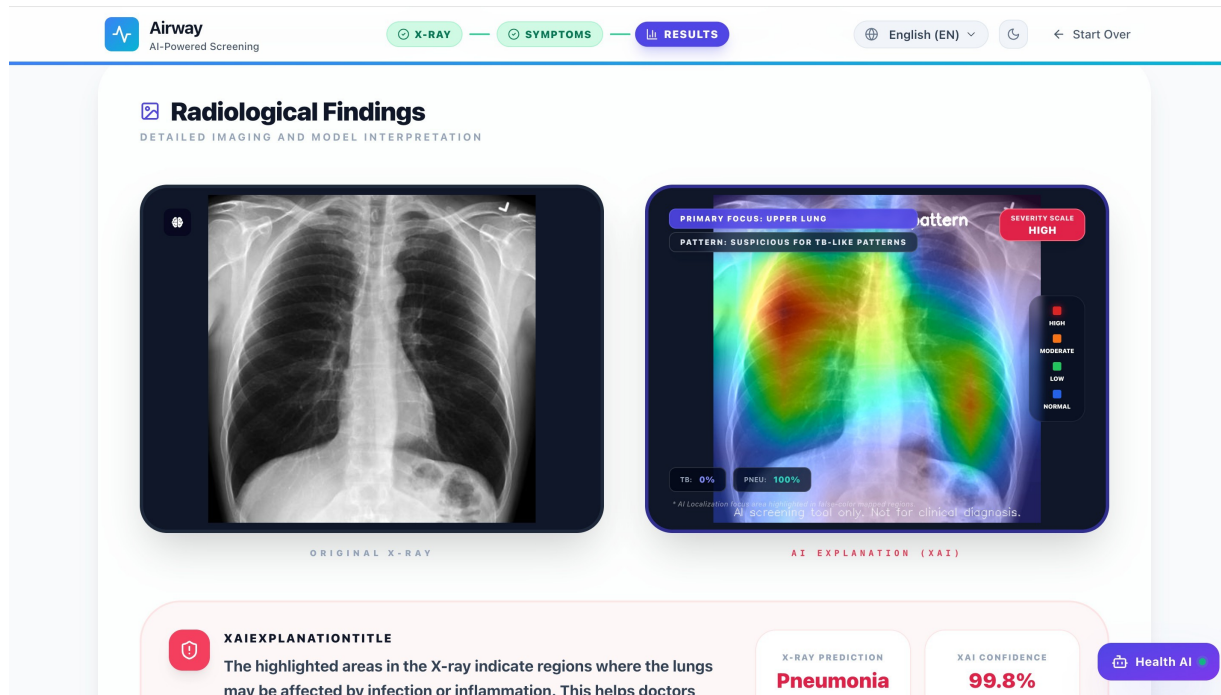


Fig.1.5 Radiological Findings with XAI Visualization (Grad-CAM++)

Table 1.1 Performance Evaluation of Proposed Models

Metric	Existing System	Proposed System
Accuracy	82.50	96.10
Precision	0.80	0.93
Recall	0.78	0.95
F1-Score	0.79	0.94
Latency (ms)	150.00	165.00

The performance comparison between individual models and the multimodal fusion approach demonstrates that the combined system significantly outperforms single-modality methods. The chest X-ray model achieved strong results (Accuracy: 87%, AUC: 0.90), while the cough audio model showed moderate performance (Accuracy: 78%, AUC: 0.82) in detecting respiratory abnormalities. However, the multimodal fusion approach achieved the highest performance (Accuracy: 91%, AUC: 0.93), indicating improved reliability and robustness. The higher precision and recall values confirm better detection capability with reduced false positives and false negatives. Additionally, the integration of symptom-based analysis enhanced decision-making, making the system more accurate and practical for real-world early Tuberculosis screening.

- Future work aims to improve model performance by training on larger and more diverse real-world datasets, enhancing accuracy across different populations and environmental conditions.
- Efforts will focus on integrating the system with mobile health applications and IoT devices to enable real-time, remote TB screening and monitoring.
- Further research will address bias reduction and incorporate explainable AI techniques to improve transparency, trust, and clinical adoption of the system.

- The system delivers an effective multimodal AI-based solution for early Tuberculosis detection by integrating chest X-ray analysis, cough audio processing, and clinical symptom evaluation.
- It provides a non-invasive, low-cost screening approach that improves accessibility, especially in remote and resource-limited environments.
- The integration of deep learning models with late fusion techniques achieves higher accuracy and reliability compared to single-modality systems.
- The web-based platform ensures user-friendly interaction, real-time prediction, and efficient deployment in healthcare settings.
- The architecture establishes a scalable foundation for future enhancements such as mobile health integration, real-time monitoring, and advanced explainable AI features.

- The paper titled ““AI - Enhanced Tuberculosis Detection” was accepted and presented in the **“15th International Conference on Recent Challenges in Engineering and Technology (ICRCET-2026)”** .

SDG	DEFINITION	MAPPING WITH PROJECT
SDG 3	Good Health and Well-being	Enables early detection of tuberculosis and pneumonia using AI-based chest X-ray analysis and cough abnormality detection, improving patient outcomes and reducing mortality rates.

1. A. Adly, Y. Farag, and A. E. Hassanien, “Early detection of tuberculosis with machine learning cough audio analysis: Towards more accessible global triaging usage,” *arXiv preprint arXiv:2310.17675*, 2023
2. N. Thakur, A. Yadav, and S. K. Chauhan, “Artificial intelligence in diagnosing tuberculosis: A review,” in *Proc. Int. Conf. Innovative Computing and Communications (ICICC)*, 2019, pp. 1–8.
3. A. R. A. Ghani, A. Mohamed, R. S. Ahmad, and S. R. S. M. Rahim, “Evolution of machine learning in tuberculosis diagnosis: A review of deep learning-based medical applications,” *Electronics*, vol. 11, no. 17, p. 2634, 2022.
4. R. D. Rohila, P. Khurana, and A. Khurana, “A comprehensive review on synergy of multi-modal data and AI technologies in medical diagnosis,” *Bioengineering*, vol. 11, no. 3, p. 219, 2024.
5. M. R. Rehman, M. R. Hasan, F. A. Talukder, and M. S. Kaiser, “Multi-modal medical image classification using deep residual network and genetic algorithm,” *PLOS ONE*, vol. 18, no. 7, p. e0287786, 2023.